

Boundary Localization Quotients and Induced Modules

$$M_{0,0}(\psi) \cong T_{f_1} M_{-1,1}(\varphi)$$

Progress report

September 7, 2026

1 Definitions, notation, and assumptions

All vector spaces and tensor products are over $\mathbb{C}$. We use the affine Lie algebra without the degree derivation:

$$\widehat{\mathfrak{sl}}_2 = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K, \quad U = U(\widehat{\mathfrak{sl}}_2).$$

Write $x_i = x \otimes t^i$. For $i, j \in \mathbb{Z}$, the brackets are

$$\begin{aligned} [e_i, f_j] &= h_{i+j} + i\delta_{i+j,0}K, & [h_i, h_j] &= 2i\delta_{i+j,0}K, \\ [h_i, e_j] &= 2e_{i+j}, & [h_i, f_j] &= -2f_{i+j}, \end{aligned} \tag{1}$$

with K central and $[e_i, e_j] = [f_i, f_j] = 0$; δ denotes the Kronecker delta.

Induced modules. Fix $a, b \in \mathbb{Z}$ with $a + b \geq 0$, and put $L = a + b + 1 \geq 1$. Set

$$S_{a,b} = \text{span}_{\mathbb{C}}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

The brackets show that this is a Lie subalgebra, with

$$[S_{a,b}, S_{a,b}] = \text{span}_{\mathbb{C}}\{e_i \ (i > a), f_j \ (j > b), h_k \ (k \geq L + 1)\}.$$

Thus a character $\chi : S_{a,b} \rightarrow \mathbb{C}$ is specified by arbitrary $c, \lambda_0, \dots, \lambda_L \in \mathbb{C}$, through

$$\chi(K) = c, \quad \chi(h_i) = \lambda_i \ (0 \leq i \leq L), \quad \chi(h_i) = 0 \ (i > L), \quad \chi(e_i) = \chi(f_j) = 0,$$

where the root vectors range over those in $S_{a,b}$. Set $\lambda_i = 0$ for $i > L$. Define

$$\mathbb{C}_\chi = \mathbb{C}1_\chi, \quad M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_\chi, \quad u_\chi = 1 \otimes 1_\chi.$$

Here $s1_\chi = \chi(s)1_\chi$ for $s \in S_{a,b}$. In particular, $h_0 \in S_{a,b}$, so $h_0 u_\chi = \lambda_0 u_\chi$. These conventions agree with [1, §2.3].

PBW notation. Let Γ_r be the set of finite nondecreasing integer sequences $\alpha = (i_1, \dots, i_k)$ with $i_\nu \leq r$, including the empty sequence. Write

$$\ell(\alpha) = k, \quad x(\alpha) = x_{i_1} \cdots x_{i_k}, \quad x(\emptyset) = 1.$$

Choose any total order on a basis of a vector-space complement of $S_{a,b}$, followed by a basis of $S_{a,b}$. PBW gives a basis of the induced module by the corresponding ordered complement monomials applied to u_χ . The complement need not be a Lie subalgebra.

Localization and its quotient. For the boundary element $F = f_b$, let $\Sigma_F = \{F^r : r \geq 0\}$. The derivation $\text{ad } F$ is locally nilpotent on $\widehat{\mathfrak{sl}}_2$, hence on U . The finite commutation formulas imply the left and right Ore conditions, so set

$$U_F = \Sigma_F^{-1}U, \quad \mathcal{D}_F M = U_F \otimes_U M, \quad T_F M = \mathcal{D}_F M / M.$$

The quotient notation assumes that F acts injectively on M , which we verify below by PBW. Since M is a U -submodule of $\mathcal{D}_F M$, the quotient is a U -module. The expression $F^{-m}u + M$ always denotes a coset of a vector in the localization; it does not assert that F^{-1} acts on the quotient. A module V is smooth if each $v \in V$ is annihilated by e_r, h_r, f_r for all sufficiently large r .

The basic case. Fix

$$M = M_{-1,1}(\varphi), \quad u = u_\varphi, \quad F = f_1, \quad T = T_F M, \quad w_m = F^{-m}u + M \quad (m \geq 1),$$

and write $c = \varphi(K)$, $\lambda_i = \varphi(h_i)$ for $i \geq 0$. Then

$$\begin{aligned} e_i u &= 0 \quad (i \geq 0), & f_j u &= 0 \quad (j \geq 2), \\ h_i u &= \lambda_i u \quad (i \geq 0), & K u &= c u, & \lambda_i &= 0 \quad (i \geq 2). \end{aligned} \tag{2}$$

Define $\psi : S_{0,0} \rightarrow \mathbb{C}$ by

$$\begin{aligned} \psi(K) &= c, & \psi(h_0) &= \lambda_0 + 2, & \psi(h_1) &= \lambda_1, & \psi(h_i) &= 0 \quad (i \geq 2), \\ \psi(e_i) &= \psi(f_i) = 0 & & & & & (i \geq 1), \end{aligned}$$

and put $A = M_{0,0}(\psi)$, $v = 1 \otimes 1_\psi$. The isomorphism requires $\lambda_1 \neq 0$; the level c is arbitrary. The additional condition $c \neq -2$ is used only for simplicity.

For common PBW indices on the source and target, set

$$\mathcal{X} = \{X = e(\alpha)h(\beta)f(\gamma) : \alpha, \beta \in \Gamma_{-1}, \gamma \in \Gamma_0\}, \quad d(X) = 2\ell(\alpha) + \ell(\beta).$$

Write $B_{m,X} = F^{-m}Xu + M$, and define vector subspaces

$$T_{\leq q} = \text{span}_{\mathbb{C}}\{B_{m,X} : m \geq 1, d(X) \leq q\} \quad (q \geq 0), \quad T_{\leq q} = 0 \quad (q < 0). \tag{3}$$

This is a vector-space filtration; its terms need not be U -submodules.

2 The basic isomorphism

Theorem 1. *If $\lambda_1 \neq 0$, the assignment*

$$\Phi : A = M_{0,0}(\psi) \longrightarrow T_{f_1} M_{-1,1}(\varphi), \quad \Phi(xv) = xw_1 \quad (x \in U)$$

is the unique $\widehat{\mathfrak{sl}}_2$ -module isomorphism sending v to w_1 . This holds for every $c \in \mathbb{C}$.

2.1 Bases and commutation with negative powers

Put F first in the complement PBW order for M . Then

$$\{F^r Xu : r \geq 0, X \in \mathcal{X}\}$$

is a basis of M . Consequently M is free over $\mathbb{C}[F]$, and F acts injectively. The localization has basis $\{F^r Xu : r \in \mathbb{Z}, X \in \mathcal{X}\}$: common denominators give spanning, and multiplying a finite

relation on the left by a sufficiently large power of F reduces independence to PBW in M . It follows that

$$\begin{aligned} \{B_{m,X} : m \geq 1, X \in \mathcal{X}\} & \text{ is a basis of } T, \\ \{Xe_0^p v : p \geq 0, X \in \mathcal{X}\} & \text{ is a basis of } A. \end{aligned} \quad (4)$$

For the second basis, place e_0 after all factors of X and before the inducing subalgebra. In particular, $w_1 \neq 0$.

Put $D = \text{ad } F$. The brackets give

$$\begin{aligned} D(f_n) &= 0, & D(h_n) &= 2f_{n+1}, \\ D(e_n) &= -(h_{n+1} + n\delta_{n+1,0}K), & D^2(e_n) &= -2f_{n+2}, & D^3(e_n) &= 0. \end{aligned} \quad (5)$$

For $x \in U$, $m \geq 1$, the following finite sum is an identity in U_F :

$$xF^{-m} = \sum_{j \geq 0} \binom{m+j-1}{j} F^{-m-j} D^j(x). \quad (6)$$

For $m = 1$, multiply the right-hand side on the right by F and use $D^j(x)F = FD^j(x) - D^{j+1}(x)$; the sum telescopes to x . Iteration gives the formula for general m , using $\sum_{r=0}^j \binom{m+r-1}{r} = \binom{m+j}{j}$.

2.2 Well-definedness and the module homomorphism

Equation (6) gives

$$\begin{aligned} f_n F^{-m} &= F^{-m} f_n, \\ h_n F^{-m} &= F^{-m} h_n + 2m F^{-m-1} f_{n+1}, \\ e_n F^{-m} &= F^{-m} e_n - m F^{-m-1} (h_{n+1} + n\delta_{n+1,0}K) - m(m+1) F^{-m-2} f_{n+2}. \end{aligned} \quad (7)$$

Taking $m = 1$ and applying (2) yields

$$\begin{aligned} e_n w_1 &= f_n w_1 = 0 \quad (n \geq 1), & K w_1 &= c w_1, \\ h_0 w_1 &= (\lambda_0 + 2) w_1, & h_1 w_1 &= \lambda_1 w_1, & h_n w_1 &= 0 \quad (n \geq 2). \end{aligned}$$

Indeed, $f_1 w_1 = u + M = 0$. For $n \geq 1$, the correction terms for e_n vanish because $h_{n+1} u = f_{n+2} u = 0$; those for h_n vanish because $f_{n+1} u = 0$. The correction for h_0 is $2F^{-2} f_1 u + M = 2w_1$. Thus $sw_1 = \psi(s)w_1$ for all $s \in S_{0,0}$.

Let $\psi_U : U(S_{0,0}) \rightarrow \mathbb{C}$ be the unital algebra character extending ψ . Applying the preceding relation to finite products and linear combinations gives $aw_1 = \psi_U(a)w_1$ for every $a \in U(S_{0,0})$. The linear map

$$U \otimes_{\mathbb{C}} \mathbb{C}_\psi \longrightarrow T, \quad x \otimes z1_\psi \longmapsto z x w_1$$

kills every balancing relation, since

$$xa \otimes z1_\psi - x \otimes a(z1_\psi) \longmapsto z x (aw_1 - \psi_U(a)w_1) = 0.$$

It therefore descends to a well-defined $\Phi : A \rightarrow T$. For $y, x \in U$, $\Phi(y(x \otimes 1_\psi)) = y x w_1 = y \Phi(x \otimes 1_\psi)$, so Φ is U -linear. Uniqueness follows from $A = Uv$. No nonvanishing assumption on λ_1 has been used at this stage.

2.3 Explicit preimages of all pure negative powers

Taking $n = 0$ in (7), and using $e_0 u = f_2 u = 0$, $h_1 u = \lambda_1 u$, gives

$$e_0 w_m = -m \lambda_1 w_{m+1}, \quad e_0^p w_1 = (-1)^p p! \lambda_1^p w_{p+1} \quad (p \geq 0). \quad (8)$$

Since $\lambda_1 \neq 0$, we have

$$w_m = \Phi(q_m), \quad q_m = \frac{(-1)^{m-1}}{(m-1)! \lambda_1^{m-1}} e_0^{m-1} v \quad (m \geq 1). \quad (9)$$

This explicitly gives a preimage of each $f_1^{-m} u + M$, without assuming surjectivity.

2.4 A strictly decreasing integer

Lemma 2. For $m \geq 1$ and $X \in \mathcal{X}$,

$$Xw_m = B_{m,X} + R_{m,X}, \quad R_{m,X} \in T_{\leq d(X)-1}, \quad (10)$$

where $R_{m,X}$ is a finite linear combination.

Proof. Assign nonnegative weights

$$\text{wt}(e_i) = 2, \quad \text{wt}(h_i) = 1, \quad \text{wt}(f_i) = 0, \quad \text{wt}(K) = 0,$$

and filter U by total word weight, writing $U_{\leq q} = 0$ for $q < 0$. Each bracket has weight at most the sum of its input weights: for $[h, e]$, $[e, f]$, $[h, f]$, and $[h, h]$, the output weights are at most 2, 1, 0, 0, whereas the input sums are 3, 2, 1, 2. Thus PBW reordering never increases weight. Neither does evaluating the inducing factors on u , where they become scalars or zero.

Equation (5) and the Leibniz rule give

$$D(U_{\leq q}) \subset U_{\leq q-1}, \quad D^j(X) \in U_{\leq d(X)-j}.$$

In particular $D^j(X) = 0$ for $j > d(X)$. Expanding in the PBW basis of M , we obtain a finite sum

$$D^j(X)u = \sum_{r \geq 0, X' \in \mathcal{X}} c_{j,r,X'} F^r X' u, \quad c_{j,r,X'} \neq 0 \implies d(X') \leq d(X) - j.$$

Substitute this into (6). The $j = 0$ term is exactly $B_{m,X}$. For $j \geq 1$, a term with $r \geq m + j$ belongs to M and vanishes in T . Every surviving term is $B_{m+j-r,X'}$, with $d(X') < d(X)$. This proves (10). $\square$

2.5 Surjectivity by strong induction

For $d \geq 0$, prove the statement

$$P(d): \quad B_{m,X} \in \text{Im } \Phi \quad \text{for every } m \geq 1 \text{ and every } X \in \mathcal{X} \text{ with } d(X) \leq d.$$

The induction hypothesis includes all denominator exponents simultaneously; an increase in the denominator exponent of a correction term is therefore harmless.

For $d = 0$, we have $X = f(\gamma)$, which commutes with F . Hence

$$B_{m,X} = Xw_m = \Phi(Xq_m).$$

The base case includes every f -monomial, not just $X = 1$.

Suppose $d \geq 1$ and $P(d-1)$ holds. Fix $m \geq 1$ and $d(X) = d$. Lemma 2 gives a finite expansion

$$\Phi(Xq_m) = B_{m,X} + \sum_{\nu=1}^s c_\nu B_{m_\nu, X_\nu}, \quad m_\nu \geq 1, \quad d(X_\nu) \leq d-1.$$

By induction, choose $a_\nu \in A$ with $\Phi(a_\nu) = B_{m_\nu, X_\nu}$. Then

$$B_{m,X} = \Phi \left(Xq_m - \sum_{\nu=1}^s c_\nu a_\nu \right).$$

This proves $P(d)$ and recursively constructs a preimage. Each recursion involves finitely many terms and strictly decreases the nonnegative integer d , so it terminates. The target basis in (4) proves surjectivity.

For example, $d(h_{-1}) = 1$, and

$$h_{-1}w_m = B_{m,h_{-1}} + 2mB_{m+1,f_0}.$$

The correction has degree zero, giving the concrete preimage

$$B_{m,h_{-1}} = \Phi(h_{-1}q_m - 2mf_0q_{m+1}).$$

This also illustrates why the induction must include all m .

2.6 Injectivity by the highest degree

By (8) and Lemma 2,

$$\Phi(Xe_0^p v) \equiv (-1)^p p! \lambda_1^p B_{p+1,X} \pmod{T_{\leq d(X)-1}}. \quad (11)$$

Let $0 \neq a = \sum_{X,p} a_{X,p} X e_0^p v \in A$, a finite sum, and let $d = \max\{d(X) : a_{X,p} \neq 0\}$. In $T_{\leq d}/T_{\leq d-1}$,

$$\Phi(a) + T_{\leq d-1} = \sum_{\substack{X,p \\ d(X)=d}} a_{X,p} (-1)^p p! \lambda_1^p (B_{p+1,X} + T_{\leq d-1}).$$

These cosets are linearly independent by (4), and $(-1)^p p! \lambda_1^p \neq 0$. Thus $\Phi(a) \neq 0$, proving injectivity and Theorem 1. Only finite linear combinations are used; the filtration terms need not be finite-dimensional.

3 Direct extensions

3.1 The boundary of $M_{-1,N}$

Theorem 3. *Let $N \geq 1$, $M = M_{-1,N}(\varphi)$, $u = u_\varphi$, $c = \varphi(K)$, and $\lambda_i = \varphi(h_i)$. Assume $\lambda_N \neq 0$. Define $\psi : S_{0,N-1} \rightarrow \mathbb{C}$ by*

$$\begin{aligned} \psi(K) &= c, & \psi(h_i) &= \lambda_i + 2\delta_{i,0} \quad (i \geq 0), \\ \psi(e_i) &= 0 \quad (i \geq 1), & \psi(f_j) &= 0 \quad (j \geq N). \end{aligned}$$

Then there is an isomorphism

$$\Phi_N : M_{0,N-1}(\psi) \xrightarrow{\sim} T_{f_N} M_{-1,N}(\varphi), \quad v \mapsto f_N^{-1} u + M.$$

Proof. Put $F = f_N$, $w_m = F^{-m} u + M$. In the commutation formulas, replace $n+1, n+2$ by $n+N, n+2N$, respectively. Now $e_i u = 0$ for $i \geq 0$, $f_j u = 0$ for $j > N$, and $\lambda_i = 0$ for $i > N$. Consequently $e_i w_1 = 0$ for $i \geq 1$ and $f_j w_1 = 0$ for $j \geq N$; at the boundary $f_N w_1 = u + M = 0$. Also

$$h_i w_1 = (\lambda_i + 2\delta_{i,0}) w_1 \quad (i \geq 0), \quad K w_1 = c w_1.$$

These relations give the well-defined homomorphism Φ_N by the balancing argument. Since $e_0 u = f_{2N} u = 0$,

$$e_0 w_m = -m \lambda_N w_{m+1}, \quad w_m = \Phi_N \left(\frac{(-1)^{m-1} e_0^{m-1} v}{(m-1)! \lambda_N^{m-1}} \right).$$

Replace $\mathcal{X}$ by

$$\mathcal{X}_N = \{e(\alpha)h(\beta)f(\gamma) : \alpha, \beta \in \Gamma_{-1}, \gamma \in \Gamma_{N-1}\}.$$

The target and source bases are $\{F^{-m} X u + M\}$ and $\{X e_0^p v\}$. The same degree $d(X) = 2\ell(\alpha) + \ell(\beta)$ is strictly lowered by $\text{ad } f_N$, while reordering and inducing relations do not increase it. Thus Lemma 2 applies unchanged. The same strong induction proves surjectivity, and the highest-degree coefficients $(-1)^p p! \lambda_N^p \neq 0$ prove injectivity. $\square$

3.2 A single step at an arbitrary boundary

Corollary 4. *Let $a, b \in \mathbb{Z}$, $a+b \geq 0$, $L = a+b+1$, $M = M_{a,b}(\chi)$, and $u = u_\chi$. If $\lambda_L = \chi(h_L) \neq 0$, then*

$$M_{a+1,b-1}(\chi^+) \xrightarrow{\sim} T_{f_b} M, \quad v \mapsto f_b^{-1} u + M,$$

where χ^+ vanishes on all root vectors in $S_{a+1,b-1}$, and

$$\chi^+(K) = c, \quad \chi^+(h_i) = \lambda_i + 2\delta_{i,0} \quad (i \geq 0).$$

Proof. Set $F = f_b$, $Y = e_{a+1}$, $w_m = F^{-m}u + M$. For $i \geq a + 2$, commuting e_i past F^{-1} gives a Cartan index $i + b \geq L + 1$, so the Cartan and central terms vanish on u . The remaining correction vanishes because $i + 2b > b$, hence $f_{i+2b}u = 0$. The other inducing relations are

$$f_j w_1 = 0 \ (j \geq b), \quad h_i w_1 = (\lambda_i + 2\delta_{i,0})w_1 \ (i \geq 0), \quad K w_1 = c w_1.$$

Thus the homomorphism is well defined. Since $Y u = 0$, $L > 0$, and $f_{a+1+2b}u = 0$, we obtain $Y w_m = -m\lambda_L w_{m+1}$. Use

$$\mathcal{X}_{a,b} = \{e(\alpha)h(\beta)f(\gamma) : \alpha \in \Gamma_a, \beta \in \Gamma_{-1}, \gamma \in \Gamma_{b-1}\}$$

and the source basis $\{XY^p v\}$. The derivation $\text{ad } f_b$ strictly lowers $d(X) = 2\ell(\alpha) + \ell(\beta)$, so the same surjectivity induction and highest-degree comparison apply, with coefficient $(-1)^p p! \lambda_L^p \neq 0$. $\square$

3.3 Smoothness, simplicity, and the assumptions

The induced modules above are smooth. Indeed, move sufficiently high modes past a fixed finite PBW word; all resulting noncentral modes still have sufficiently high indices and annihilate the inducing vector. The bound depends on the word. The isomorphisms therefore also establish smoothness of the localization quotients.

By [1, Theorem 1.1], for $a + b \geq 0$,

$$M_{a,b}(\chi) \text{ is simple} \iff \lambda_{a+b+1} \neq 0 \text{ and } c \neq -2.$$

Under the hypothesis $\lambda_L \neq 0$ of Corollary 4, the boundary quotient is simple if and only if $c \neq -2$. The isomorphism remains valid at the critical level $c = -2$; its injectivity was proved without using simplicity.

If $\lambda_1 = 0$, the basic homomorphism still exists, but $\Phi(e_0 v) = e_0 w_1 = 0$, whereas $e_0 v \neq 0$ by the source PBW basis. Thus the specified map is not an isomorphism.

References

- [1] V. Futorny, X. Guo, Y. Xue and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2 (2024), §2.3 and Theorem 1.1. <https://arxiv.org/abs/2404.03855v2>.